CRF and Cl(3): Symmetry, Not Dynamics

Clifford Charge, Grade Current, and the Fourth Interpretation of $f_{\mathrm{II,crit}}$

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Companion to CRF Clifford (v1–v2). Resolves the Born-rule question and develops the grade current equation.

Abstract. Three results extend the CRF–Clifford connection established in the companion documents.

(1) Born rule is not Clifford multiplication (proved). The Born rule $E[P_{\mathrm{new}}] = P_{\mathrm{avg}}$ is linear in P. The Clifford product is bilinear/nonlinear. These cannot be the same operation. The correct framing: Cl(3) is the *symmetry group* of the CRF mode space, not the operation of the Born rule.

(2) Grade current equation. The Clifford charge $C(P_i) = \sum_k P_{i,k} \cdot \mathrm{sign}(e_k^2)$ satisfies a reaction-diffusion equation:

$$\frac{dC_i}{dt} = \sum_j W_{ij}(C_j - C_i) + S_i(t)$$

where W_{ij} is the CRF coupling (proven grade-preserving) and $S_i = +1$ at crystallisation events. Total charge $Q = \sum_i C_i$ is conserved by the diffusion term alone.

(3) Fourth interpretation of $f_{\mathrm{II,crit}}$. $f_{\mathrm{II,crit}} = 0.14700$ is the *critical Clifford charge density* for the d_s-lock transition. It is the minimum fraction of space-charged ($C = +1$) nodes required for the grade-1 subgraph to percolate and lock $d_s = 3$. This is the fourth independent reading of the same number (alongside percolation threshold, crossing probability, and Phase II fraction).

Born Rule is Not Clifford Multiplication

Theorem 1 (Linearity argument)**.** *The Born rule cannot implement Clifford multiplication.*

Proof. *The Born rule:* $E[P_{\mathrm{new}}] = P_{\mathrm{avg}} = \frac{1}{k}\sum_j P_j$*. This is a* linear *function of the input distributions.*

The Clifford product of two state vectors $|\psi_A\rangle = \sum_i P_{A,i}\, e_i$ *and* $|\psi_B\rangle = \sum_j P_{B,j}\, e_j$*:*

$$|\psi_A\rangle \cdot |\psi_B\rangle = \sum_{i,j} P_{A,i}\, P_{B,j}\, \mathrm{sign}(i,j)\, e_{i\oplus j}$$

is bilinear *(quadratic) in* (P_A, P_B)*.*

A linear operation cannot equal a bilinear one for all input distributions. □

Numerical verification (case $|\{x\}\rangle \cdot |\{y\}\rangle$ vs. average($\{x\}, \{y\}$)):

- Clifford product: all weight on mode $\{xy\}$ (gauge, grade 2)
- Born average: weight 1/2 on each of $\{x\}$ and $\{y\}$ (grade 1)

These are completely different.

Correct framing

Cl(3) plays the role of the *symmetry group* of the CRF mode space, not the operation of the Born rule. This parallels quantum mechanics: SU(2) is the symmetry group of spin states, but the Schrödinger equation is a U(1) operation in time. Both the Cl(3) symmetry and the Born rule coexist in CRF:

	Cl(3) symmetry	Born rule
Role	Static structure	Dynamic operation
Type	Algebraic (Clifford)	Stochastic (Dirichlet)
Grade effect	Preserves grade	Averages across grades
Conserves	Clifford charge C	Probability norm

The Grade Current Equation

Clifford charge

For a node with probability distribution $P = (P_0, \ldots, P_7)$, define the *Clifford charge*:

$$C(P) = \sum_{i=0}^{7} P_i \cdot \mathrm{sign}(e_i^2) \;=\; P(\text{grade 0,1}) - P(\text{grade 2,3})$$

The grade-sign pattern in Cl(3, 0): grades 0 and 1 have $e^2 = +1$ (positive); grades 2 and 3 have $e^2 = -1$ (negative). Counts are equal ($4 + 4 = 8$), so the ground state (uniform P) has $C = 0$.

Reaction-diffusion dynamics

Grade current equation — from Born rule + crystallisation

$$\frac{dC_i}{dt} = \underbrace{\sum_j W_{ij}(C_j - C_i)}_{\text{diffusion}} + \underbrace{S_i(t)}_{\text{source}}$$

Diffusion term. From $E[C_{\text{new}}] = C_{\text{avg}}$ (Born rule conserves C in expectation) and W_{ij} being grade-preserving (CRF Clifford doc.), the mean charge at node i evolves as a weighted average of neighbors.
Source term. $S_i(t) = +1$ when node i crystallises (Phase I $\to$ Phase II), $S_i(t) = 0$ otherwise. Crystallisation is a Clifford charge creation event.
Conservation. $\sum_i dC_i/dt\big|_{\text{diffusion}} = 0$ (antisymmetric sum over symmetric W_{ij}). Total charge $Q = \sum_i C_i$ changes only via crystallisation sources.

Grade current

The *grade current* between nodes i and j:

$$J_{ij} = W_{ij}(C_j - C_i) = -J_{ji}$$

This is a conserved antisymmetric current analogous to diffusion current $\mathbf{J} = -D\nabla C$ in continuum physics. In the CRF graph, it flows from high-charge regions to low-charge regions, driven by the W_{ij} coupling.

Fourth Interpretation of $f_{\text{II,crit}}$

Four interpretations — all give $f_{\text{II,crit}} = 0.14700$

1. **Percolation threshold.** Minimum fraction for the 3D vector-Ω subgraph to span the network.
2. **Crossing probability.** $P(N \geq 1 \,|\, \text{Poisson}(\sqrt{\varphi}/n))$: probability a node crosses the mind threshold in one T_{avg}.
3. **Phase II fraction.** Steady-state fraction of crystallised nodes (observed invariant in V20.3 at both 100 and 400 sweeps).
4. **Critical Clifford charge density (new).** The minimum Clifford charge density $\bar{C} = (\sum_i C_i)/N = f_{\text{II,crit}}$ required for the grade-1 (space) modes to percolate and lock $d_s = 3$.

The fourth interpretation follows from the grade current equation: the charge density $\bar{C}$ must reach a critical value before the grade-1 sector can form a spanning cluster (the d_s-lock percolation condition). This critical value is $f_{\mathrm{II,crit}}$.

Note on kinetic equilibrium. Long-time kinetics (HN-02) give $f_{\mathrm{II}} \to 1$ as $t \to \infty$. The value $f_{\mathrm{II,crit}} = 0.147$ is the charge density at the *moment of* d_s-lock, not the long-time equilibrium. It is a topological threshold, not a kinetic steady state.

Testable Prediction: 5 Psi-Clusters = 5 Grade Sectors

Cluster–grade correspondence (V18 test)

The 5 Psi-clusters detected in V18 simulation (N=500, 400 sweeps) should correspond to the 5 non-space grade sectors of Cl(3):

Cluster	Grade sector	Cl(3) basis	Ω-space signature
$\Psi = -2\Delta_\Psi$	grade 3 $\{xyz\}$	pseudoscalar	all 3 components
$\Psi = -\Delta_\Psi$	grade 2 $\{yz\}$	e_2e_3	$\Omega_x \approx 0$
$\Psi = 0$	grade 2 $\{xz\}$	e_1e_3	$\Omega_y \approx 0$
$\Psi = +\Delta_\Psi$	grade 2 $\{xy\}$	e_1e_2	$\Omega_z \approx 0$
$\Psi = +2\Delta_\Psi$	grade 0 $\{\}$	scalar	$\lvert\Omega\rvert \approx 0$

where $\Delta_\Psi = 2z_N\sigma_\Psi/(n-1) = 0.01529$ (for $N = 500$).

Key signatures: (a) Outermost clusters ($\pm 2\Delta_\Psi$) = grade-0 and grade-3 (SO(3) singlets); (b) Inner clusters ($0, \pm\Delta_\Psi$) = grade-2 triplet; (c) Each grade-2 cluster has Ω concentrated in one coordinate plane.

Test: extract per-cluster Ω histograms from V18 data and check whether the cluster at $\Psi = 0$ has $\lvert\Omega_y\rvert \approx 0$ (or similar systematic pattern). If confirmed, this would be the first *direct* evidence of Clifford grade structure in CRF simulation.

Status

Result	Status	Basis
Born rule $\neq$ Clifford product	**Proved**	linearity argument
Cl(3) = symmetry, not dynamics	Derived	from proof above
Grade current equation	Derived	from Born + W_{ij}
Total charge Q conserved (diffusion)	**Proved**	antisymmetry of W_{ij}
$f_{\mathrm{II,crit}}$ = 4th interpretation	Hypothesis	grade-current framework
5 Psi-clusters = 5 grade sectors	Prediction	testable via V18

Summary

The Born rule is linear; the Clifford product is nonlinear. They cannot be the same operation. Cl(3) is the *symmetry group* of CRF mode space, governing which transitions are grade-allowed, not the operation that implements them.

The Clifford charge $C(P_i) = P(\text{grade } 0{,}1) - P(\text{grade } 2{,}3)$ satisfies a reaction-diffusion equation. Total charge is conserved by the W_{ij} diffusion; crystallisation events are sources.

$f_{\mathrm{II,crit}} = 0.14700$ acquires a fourth reading: the critical Clifford charge density for the d_s-lock transition. Space crystallises when enough grade-1 charge accumulates to form a spanning cluster in the Ω-proximity graph.

Prediction for V18: the 5 Psi-clusters correspond to the 5 non-space grade sectors of Cl(3), each with a characteristic Ω-space signature. The outermost clusters are the SO(3) singlets (grades 0 and 3); the inner three are the SO(3) triplet (grade-2).

The universe does not compute *with* Cl(3) at every Born event.
It *lives within* Cl(3), constrained by its grade structure,
while the Born rule moves probability among its grade sectors.